Detection of a Real Thai National ID Card in Video Based

Aniwat Keereena
Department of Computer Engineering
Chulalongkorn University
Bangkok, Thailand
6470301021@student.chula.ac.th

Sukree Sinthupinyo
Department of Computer Engineering
Chulalongkorn University
Bangkok, Thailand
sukree.s@chula.ac.th

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***Abstract*—Identity verification i s a m andatory p rocess when applying for various online services in Thailand. It requires presentation of your national ID card as evidence. The verification process requires users to capture a photo or video of their card. However, there is a risk of users attempting to counterfeit the card during this process. To mitigate this risk, printed holograms are applied to the card to prevent counterfeiting. The hologram provides a visual means of checking for authenticity. If a hologram is detected on the card, it serves as an indicator of a genuine card, facilitating the decision-making process in identity verification. T his m easure i s i mplemented t o e nhance security and prevent fraudulent activities. This research aims to apply deep learning techniques, utilizing the MobileNet SSD v2 as the backbone model, for hologram recognition on Thai national ID cards. We have developed a system specifically f or classifying holograms using grayscale images. In the training phase, the transfer learning method is employed, transferring pre-trained weights from the RGB model to initialize the grayscale model. The classification a ccuracy o n t he v alidation d ata i ndicates that the system's mAP (Mean Average Precision) at IoU (Intersection over Union) thresholds of 0.5 to 0.95 for the RGB model outperforms the grayscale model by 1.38%. However, the results for real card detection with grayscale video show an accuracy of 88.33%. These results demonstrate the success of developing a deep learning recognition system for detecting a real Thai national ID card in grayscale with high accuracy.**

***Index Terms*—Deep Learning, MobileNetSSD, Grayscale Images, National ID Card, Hologram**

I. Introduction

Thailand has increasingly transitioned to providing various services through online channels, especially during the COVID-19 pandemic, when many services have shifted to being almost entirely digital. This transition aims to prevent people from having to leave their homes for activities, providing greater convenience, safety, and savings in time and costs.

In this context, certain services require confirmation o f the recipient's identity before proceeding. These services include taking online exams, registering to use public or private sector applications, financial transactions, and more. For services that follow a Know Your Customer (KYC) approach, ensuring customer identity without face-to-face interaction (Non Face-to-Face) becomes crucial. Robust authentication methods are necessary to verify the customer's identity and prove their legitimacy, preventing corruption, counterfeiting, or the unauthorized use of others' information in transactions.

The primary principle of online practices involves verification primarily through the Thai national ID card. This card, issued by the government to Thai citizens listed in the house registration, serves as evidence of identity. Featuring holograms printed and coated as a symbolic representation, the card allows for visual inspection to prevent counterfeiting. When tilted, the holographic print reflects light, creating a distinctive rainbow shimmer. This uniqueness enhances safety, making it challenging to counterfeit.

In this paper, we propose a modified version of MobileNet SSD v2, a Convolutional Neural Network (CNN) architecture designed for hologram classification in Thai national ID cards using grayscale images. Our architecture was enhanced through transfer learning, utilizing pre-trained from the RGB model to the grayscale model. The primary focus of this research is to detect genuine Thai national ID cards from videos using deep learning techniques applied to grayscale imagery. In the first experiment, we trained both RGB and grayscale models using the same dataset to compare their performances. In the second experiment, the testing involved the grayscale model with grayscale videos, encompassing both authentic and forged Thai national ID cards created through photocopying, taking photos, and displaying them on a mobile phone. The objective is to confirm the legitimacy of the cards, contributing to their credibility and reliability.

II. Related Works

A. MobileNet SSD v2

The MobileNet SSD v2 [1] [2] [3] architecture efficiently delivers real-time results, even on mobile devices with limited computing power. It combines the MobileNet v2 architecture for feature extraction with the Single Shot Detector (SSD) architecture for object detection, enhanced by the addition of a Feature Pyramid Network (FPN) [4] Lite module.

MobileNet v2 is a lightweight CNN architecture designed for efficient computation on mobile and embedded devices. By employing depthwise separable convolutions and residual connections, it effectively reduces the number of parameters while enhancing network performance. Serving as a feature extraction component, MobileNet is tasked with identifying

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crucial features in the input image for subsequent object detection.

SSD is a widely used object detection algorithm that utilizes a set of default bounding boxes, known as anchor boxes, to predict the location and class of objects in an input image. The algorithm combines predictions from these anchor boxes using a Non-Maximum Suppression (NMS) algorithm to generate the final detection results. SSD functions in the detection phase, utilizing extracted features to identify the location of objects in the image.

FPN Lite is a lightweight version of the FPN architecture, commonly used in object detection tasks to improve accuracy. It achieves this by integrating high-resolution features from shallow layers with low-resolution features from deep layers. The FPN Lite module combines the output of the MobileNet v2 and SSD layers, providing additional contextual information for the object detection task. This integration enhances the accuracy of object detection by incorporating information from multiple scales.

B. Previous Research

B. Ay [5] addresses the challenge of hologram authenticity on security documents, such as foreign passports and Euro banknotes, by introducing a deep learning hologram verification system. This system determines whether a hologram image captured from a mobile application is genuine or fake. The research utilized four CNN architectures: VGGNet, DenseNet, ResNet, and MobileNetV2. Fake holograms include those with added noise and printed hologram photographs.

D. Soukup and R. Huber-Mörk [6] proposed creating real-time hologram detection for Euro banknotes of five denominations (EU5, EU20, EU50, EU100, and EU500) by developing a portable hardware module attached to a mobile device for capturing holograms using LED light. The study employed five CNN architectures: FC7-4096, FC7-1024, FC7-256, FC7-64, and FC7-16.

S. Gonzalez, A. Valenzuela, and J. Tapia [7] aimed to detect the tampering of chipless ID cards. Two types of experiments were conducted in accordance with the state of the art. The first experiment involves classifying real and composite (digitally tampered) ID card images in RGB format using the MobileNet architecture. The second experiment focuses on classifying real, printed, and displayed ID card images using the BasicNet architecture. Additionally, image processing is employed for feature extraction methods for verification. Three different feature extraction methods are utilized: Discrete Fourier Transform (DFT), Steganalysis Rich Model (SRM), and Error Level Analysis (ELA).

As mentioned earlier, hologram detection has been implemented in various security documents, utilizing RGB images as the dataset for training several CNNs to create an RGB model.

III. Methodology

This section describes the approach for detecting a genuine Thai national ID card, covering dataset details, image preprocessing, transfer learning, model evaluation, video preprocessing, and the process for detecting a real Thai national ID card.

A. Model Selection

Online services requiring KYC are designed for mobile phones due to the need for camera usage during operations such as taking pictures. To streamline object detection and minimize the burden on users when downloading large applications. We explored the use of a compact model. After considering various models [8], it was determined that MobileNetV2 had the smallest size of 14 MB [9], compared to MobileNetV3 Large with a model size of 21 MB [10]. Although MobileNetV3 Small has a smaller model size of 10 MB, its top 1 accuracy is 68.1%, which is lower than MobileNetV2's accuracy of 71.3%. Consequently, MobileNetV2 was selected for this research.

B. Dataset and Preprocessing

The dataset used for training the model was derived from legally obtained real Thai national ID cards, comprising 2,220 images that included 8,480 hologram objects. The original resolution of the images was 4032x3024 pixels, resized, and cropped to 600x600 pixels. The images were split into training 80%, testing 10%, and validation 10%. Fig 1. provides an example of the dataset before undergoing image processing.

Fig. 1. Appearances of holograms on Thai national ID cards illuminated by different reflected light sources.

Holograms, when viewed under reflected light, exhibit a color change relative to their position in the image when observed from different angles. This characteristic may impose an excessive emphasis on color information in hologram photos. Utilizing grayscale images [11] [12] can mitigate this issue by reducing color complexity, allowing deep learning to better focus on holographic features.

In Fig 2., illustrates our proposed approach to hologram classification, most deep learning models commonly use the Red, Green, and Blue (RGB) color space as an input to extract features from the three color channels. In our approach, we suggest converting RGB to Hue, Lightness, and Saturation (HLS) color space [13] [14]. This conversion aims to emphasize the reflected light of holograms in an image, making it easier to detect specific objects. Subsequently, we convert

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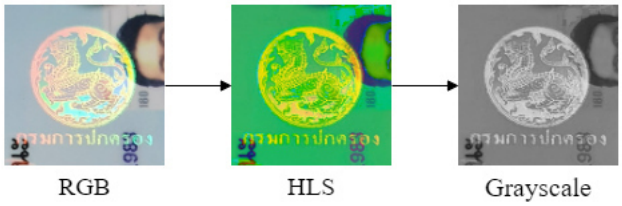

Fig. 2. The conversion process involves transforming the color space from RGB to HLS and subsequently from HLS to grayscale.

HLS to a grayscale color space, a single color channel used to extract features. In contrast to RGB and HLS color spaces, where each pixel is represented by three values, the grayscale color space represents each pixel with a single value. The grayscale value signifies the intensity of the pixel, with higher values indicating brighter pixels and lower values indicating darker pixels.

C. Training and Transfer Learning

Transfer Learning [15] is a technique that reduces the training time of a deep learning model by incorporating segments of a model already trained on similar tasks into a new model. This approach is particularly useful in situations where a large dataset is unavailable. Transfer learning serves as an initial model trained on a smaller dataset for specialized tasks, thereby aiding in the reduction of model training time. It involves integrating segments of an already trained model into a new one.

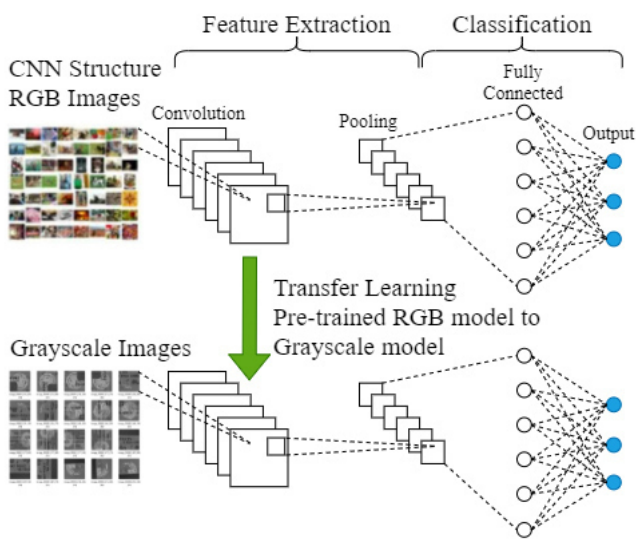

Fig. 3. Transfer Learning from pre-trained weights in the RGB model to the grayscale model.

Fig 3. Illustrates our proposed approach to customizing transfer learning from the RGB model to the grayscale model. This is necessary, as there are very few pre-trained weights available for deep learning models designed specifically for grayscale images.

D. Model Evaluation

The dataset should be divided into training data and validation data. The validation data will not be included in the training model process, even after the completion of the training. A machine learning model will predict the validation data, and the results predicted by the machine learning model will be compared with the labels of the validation data. The mAP is a metric used to measure the performance of a model. The mAP formula is based on the following sub-metrics:

- Confusion Matrix
- Intersection over Union
- Precision
- Recall

The Confusion Matrix will be employed to evaluate accuracy, precision, and recall for model comparison. It requires the following 4 components:

- TP (True Positive): The actual value was positive and the model predicted a positive value.
- FP (False Positive): Your prediction is positive, and it is false.
- TN (True Negative): The actual value was negative and the model predicted a negative value.
- FN (False Negative): Your prediction is negative, and result it is also false.

Intersection over Union is used to measure the accuracy of localization provided by an object detector on a dataset. It does this by comparing the predicted bounding box coordinates with the ground truth box coordinates to assess the degree of overlap.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

Equation (1) is Accuracy., (2) is Precision measures the quality of model by its ability to find TP out of all positive predictions. and (3) is Recall is quantity and measures the model's ability to be able to find TP out of all predictions.

$$mAP = \frac{1}{n} \sum_{i=1}^{n} AP_i \tag{4}$$

Equation (4) mAP is calculated by finding Average Precision (AP) for each class and then average over a number of classes.

- n is the number of classes.
- AP_i is Average Precision of class i.

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E. Video and Preprocessing

The video used for detecting real Thai national ID cards comprises 780 files captured from cards in Section III-B. The original video resolution was 1,440x1,440 pixels, which was resized to 600x600 pixels. The videos were categorized into real cards with 260 files, printed with 260 files, and displayed with 260 files. The video has a duration of 15 seconds with a frame rate of 30 FPS (Frames Per Second), resulting in 450 frames per video, and the audio track is disabled. Each video must capture only one national ID card.

The videos employ the same approach as the images shown in Fig 2. involving the conversion from RGB to HLS color space, followed by the transformation from HLS to a grayscale color space.

Fig. 4. Example of fake Thai national ID cards by printed and displayed on a mobile phone.

Fig 4. Illustrates our proposed approach to detecting counterfeit Thai national ID cards created through photocopying and displaying them on a mobile phone, where holograms may be present.

F. Process for Detecting a Real Thai National ID Card

The grayscale videos were processed using the grayscale model to detect holograms in every frame, totaling 450 frames. During detection, we calculated and collected information, which served as features, as shown in Table I for the Feature Selection [16] dataset to develop a real card predictive model. The dataset used for feature selection was defined in Section III-E. The videos were split into training 70% and validation 30%, utilizing Gradient Boosting [17] for classification, as the algorithm builds an additive model.

In the features, we used variance to define a measure of the spread or dispersion within a set of data. Since we only have a sample, the sample variance represents the variance of this sample. Given a sample of data of size n, the sample variance is calculated using the following formula:

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1} \tag{5}$$

TABLE I
FEATURES COLLECTED DURING THE HOLOGRAM DETECTION PROCESS.

Feature	Description
CountHoloFrame	Total count of frames in which holograms were detected.
CountHoloObj	Total count of detected hologram objects, spanning 450 frames.
MinHoloObj	Minimum number of holograms detected.
MaxHoloObj	Maximum number of holograms detected.
VariMaxHoloObj	Variance of maximum number of holograms detected.
VariMinAcc	Variance of the minimum accuracy per frame, spanning 450 frames.
VariMaxAcc	Variance of the maximum accuracy per frame, spanning 450 frames.
VariAvgAcc	Variance of the average accuracy per frame, spanning 450 frames.

Equation (4) represents the sample variance formula used as a feature.

- x_i is each individual data point.
- $\bar{x}$ is the mean (average) of the data.
- n is the number of data points. In this paper, the number of data points ranging from video frame 1 to frame 450.

IV. Experimental Results

In the first experiment, we compared the RGB and grayscale models. Both models were applied to a dataset of 222 images, using the same dataset but with a difference in color channels. The RGB model employed the three color channels dataset, while the grayscale model utilized a single color channel dataset. We trained our model using parameters from a pre-trained process. The grayscale pre-trained process involved customizing RGB weights for initial use as grayscale weights. To evaluate the models, we utilized precision, recall, and accuracy as mentioned in Section III-D.

MobileNet SSD v2 and the proposed modified model were implemented using the Python programming language based on TensorFlow. All experiments were conducted on Google Colab.

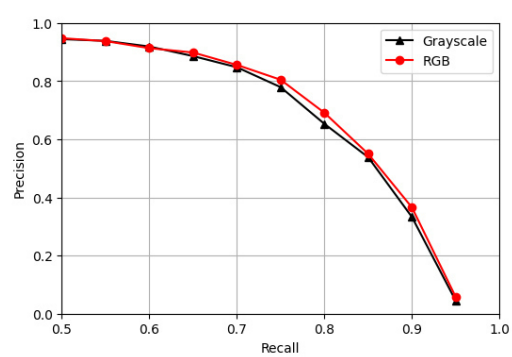

Fig. 5. The accuracy of RGB model and grayscale model at IoU 0.5:0.95.

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In Fig 5., the classification accuracy for hologram detection on the validation data 222 images shows that the system's mAP 10 IoU thresholds range from 0.5 to 0.95 with a step size of 0.05. The RGB model outperforms the grayscale model average by 1.38%.

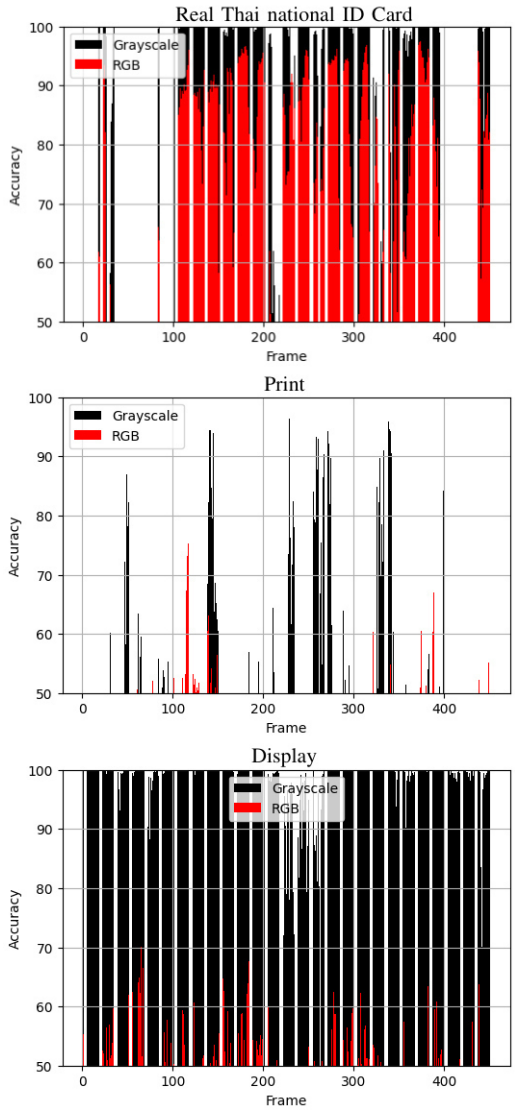

Fig. 6. An example illustrating the accuracy of hologram detection between the RGB model and the grayscale model is demonstrated across 450 frames.

Fig 6., the accuracy of hologram detection was compared between the RGB model and the grayscale model on a video frame-by-frame basis. We achieved an accuracy ratio of more than 50%. Surprisingly, the grayscale model outperformed the RGB model by approximately 29.45% for a real national ID card.

However, we can demonstrate that the grayscale model can be used for hologram detection on Thai national ID cards similarly to the RGB model, achieving very close accuracy.

In the second experiment, we conducted a performance evaluation of real Thai national ID card detection using the grayscale model. We applied the model to 180 video files. During the detection process, we calculated and collected information as shown in Table I, then generated a CSV file as data for Feature Selection. The data in the CSV file was split into 126 records for training features and 54 records for testing features. This dataset was utilized to create a model using Gradient Boosting, which was then tested with records encompassing real cards, printed cards, and displayed cards. To evaluate the model in the classification process, we identified the best features shown in Table II. The achieved accuracy is 88.33%.

TABLE II
THE BEST FEATURES AND WEIGHTS

Features	Weights
CountHoloFrame	23.85%
MaxHoloObj	28.84%
VariMaxHoloObj	26.87%
VariMinAcc	8.87%
VariMaxAcc	11.57%

When the model is tested on 600 videos, including 200 real files, 200 printed files, and 200 displayed files, the accuracy results are shown in Table III. It can be observed that it can detect Thai national ID cards with an accuracy of 82%. From the experiment, it was found that errors can be caused by various factors, such as the integrity of the national ID card, scratches on the card, validity of the card, or even the process of shooting a video. When there is insufficient light, the hologram may not reflect light, leading to more errors in forecasting.

For detecting fake ID cards, including photocopied ID cards with an accuracy of 93% and national ID cards displayed on mobile phones with an accuracy of 90%, the model demonstrated a $\leq$10% chance of making incorrect predictions distinguishing between fake and genuine ID cards. The experiment revealed that the key to successful counterfeiting lies in the quality of the image captured or displayed on the mobile phone, including sharp holographic images and a sufficient number of holograms. Additionally, movement while recording video can enhance the likelihood of successful counterfeiting.

TABLE III
PERFORMANCE OF REAL THAI NATIONAL ID CARD DETECTION

Taken From	Accuracy
Real Thai national ID Card	82%
Print (Photocopying)	93%
Display (Displaying on a mobile phone)	90%

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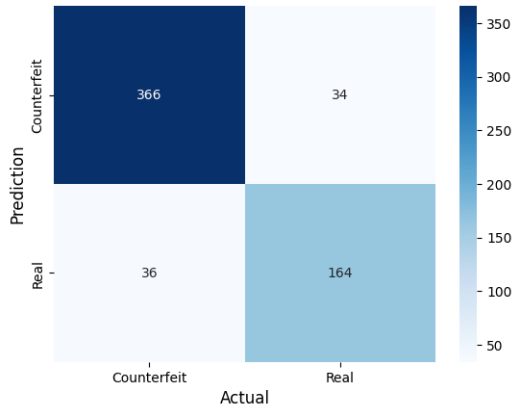

Fig. 7. The confusion matrix

Our grayscale model achieved a comparable accuracy rate to the RGB model in the first experiment. In the second experiment, the confusion matrix is shown in Fig 7. The features achieved an accuracy of 88.33% in detecting real Thai national ID cards. Thus, detecting holograms in real Thai national ID cards with grayscale color space has the potential to be further developed to identify Thai national ID cards effectively.

V. Conclusion

In this paper, we introduce our modified MobileNet SSD v2 for detecting real Thai national ID cards in videos. The results from the dataset confirmed the effectiveness of our modified model based on MobileNet SSD v2 for hologram detection with grayscale images. This solution not only helps reduce the need for preparing a large dataset for grayscale images but is also beneficial for transferring pre-trained weights from the RGB color space to the grayscale color space. Grayscale images contribute to reducing file size and network traffic. We believe this solution can be applied to other images with objects reflecting light.

Suggestions for further research are as follows: Utilize transfer learning techniques to further investigate the performance of various models. Remove limitations on model size to explore a wider range of architectures and their impact on detection accuracy. Extend the technique to detect holographic images beyond Thai national ID cards. Develop a model capable of detecting general holographic images by curating diverse datasets, including holograms on driver's licenses, credit cards, banknotes, and other documents.

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